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Simulated Nitrogen Dynamics and Losses in a Layered Agricultural Soil

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ABSTRACT

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A soil nitrogen model emphasizing mineral nitrogen dynamics and losses is presented. The model has a one-dimensional layered structure and considers plant uptake, mineralization, immobilization, leaching and denitrification processes. Fertilization and manure additions are included as management inputs. A physically based soil-water and heat model provides daily values of temperature, unfrozen water content, water flow and drainage, at different depths in the profile. Input data requirements include standard meteorological variables, basic soil physical and biological properties and crop management characteristics.

Soil nitrogen dynamics were simulated for a 3-year period in N-fertilized and unfertilized barley. Model predictions were compared with measurements of nitrate leaching and mineral N content of the soil. Simulation of mineral N levels and leaching generally agreed with field data. Prediction of mineral N dynamics in both N-fertilized and unfertilized barley was better for surface layers than for deeper layers in the profile. Discrepancies between simulated and measured mineral N content in the topsoil were mainly related to mineralization and plant uptake. In deeper soil layers, differences between measured and predicted values were primarily related to the water flow and the drainage pathway. Simulated amounts of nitrate leached were close to measured values. Discrepancies in the temporal distribution of nitrate leached were mainly attributable to the simulated water flow. The model predicted that nitrate leaching to drainage tiles occurred mainly from the upper layers. Prediction of leaching depended as much on the simulation of drainage flow pathways and the vertical distribution of nitrate in the profile as on the simulation of water flow rates.

INTRODUCTION

The fate of mineral nitrogen, whether it is retained in the crop–soil system and contributes to productivity or is transported and lost from the system, is

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an important consideration in modern agriculture. Increasing the cost–benefit of fertilizer use, in an era of rising production costs, is a major challenge (Rosswall and Paustian, 1984). The enrichment of groundwater and aquatic systems resulting from inefficient nitrogen utilization represents a serious environmental impact in many regions, including Sweden (Gustafson, 1983). Modifying agricultural practices to solve these problems requires a better quantitative understanding of nitrogen cycling processes in agricultural soils.

Simulation models provide an effective means of organizing knowledge about the complex, interactive behaviour of nitrogen in soil (Frissel and van Veen, 1982) and a number of models have been developed for a variety of purposes (reviewed by Frissel and van Veen, 1981; Haith, 1982). Models for simulating crop nitrogen availability and nitrogen losses range from complex, multi-process models (e.g. Tanji and Mehran, 1979; van Veen and Frissel, 1981; Schaffer et al., 1983) to simpler, empirically-based formulations (e.g. Addiscott, 1977; Burns, 1980; Burns and Greenwood, 1982).

Among the models which have received the widest application are simple models of solute transport and nitrate leaching. The model of Burns (1974) has been used for estimating movement of fertilizer nitrogen in several field soils (Burns, 1976; Greenwood and Burns, 1979) and for making regional estimates of nitrate leaching (Burns and Greenwood, 1982). An independent test of three leaching models was carried out by Cameron and Wild (1982) using data on chloride transport.

A general characteristic of the more complex models developed as "research tools" is that they are often difficult to apply to many different sites, due in part to extensive information requirements (Frissel and van Veen, 1982). Only a few such models have been applied to different sites or data sets (e.g. van Veen and Frissel, 1981; Parton et al., 1983), although management models (Knisel, 1980; Schaffer et al., 1983) have been developed for widespread use.

We have incorporated existing theory into a comprehensive structure of the major processes involving nitrogen in soil, with emphasis on simulating mineral nitrogen dynamics and nitrogen losses. The focus of model development was to facilitate numerous applications, with a model resolution compatible with information generally available from agricultural field research. This paper describes the model and an application at the field level to simulate temporal dynamics of soil nitrogen. Nitrogen in N-fertilized and unfertilized barley was simulated and compared with measurements over a 3-year period for the field site of the "Ecology of Arable Land" project (Persson and Rosswall, 1983; Steen et al., 1984).

MODEL DESCRIPTION

Overview of model structure

The nitrogen model (Fig. 1) includes the major processes that determine inputs, transformations and outputs of nitrogen in agricultural soils. A water

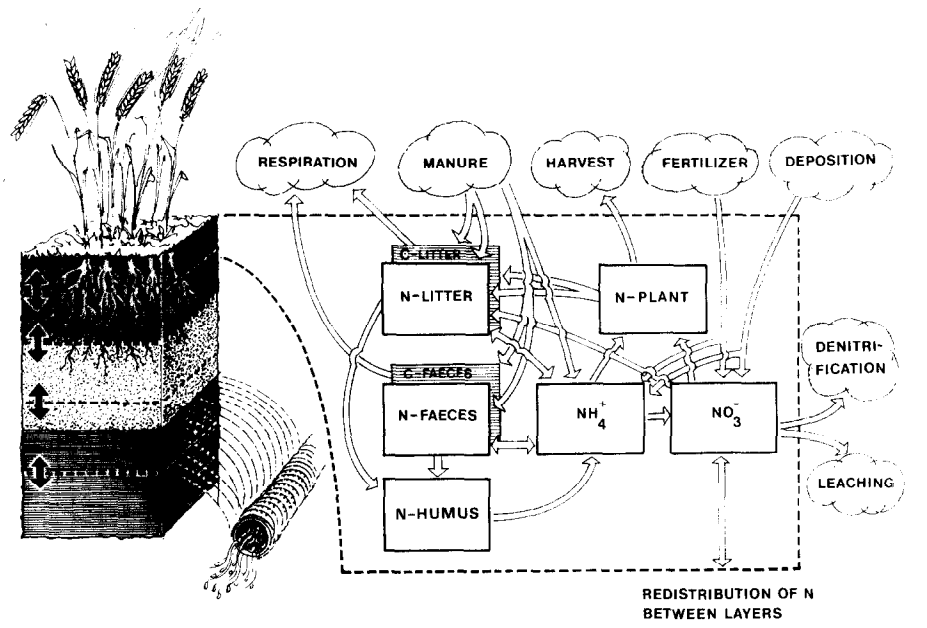


Fig. 1. The nitrogen model structure. Parts within the broken line represent the uppermost layer of the soil profile. Subsurface layers have the same structure as the surface layer but have no direct transfers from fertilizer and deposition.

and heat model (Jansson and Halldin, 1979) provides driving variables for the model, i.e. surface runoff and infiltration, water flow between soil layers and flow to drainage tiles, unfrozen soil water content and soil temperature.

Inorganic and organic pools are replicated for each soil layer. The soil profile is divided into layers (five in the present version) on the basis of physical and biological characteristics. Mineral N pools include ammonium and nitrate. Organic N is classified as litter, manure-derived faeces and humus. Carbon pools for litter and faeces are included for controlling nitrogen mineralization and immobilization rates. The litter component includes undecomposed material (e.g. crop residues, dead roots and microbial biomass), while stabilized decomposition products make up the humus component. The single plant component includes nitrogen in both above- and below-ground biomass. Root distributions within the soil profile are taken into account (see below).

Manure, inorganic fertilizer and atmospheric deposition are inputs to the uppermost soil layer (Fig. 1). Losses through denitrification and leaching of nitrate to drainage tiles can occur from each soil layer. Nitrate in solution can be transported between soil layers or to drainage tiles.

Water and heat flows

The water and heat model is based on two coupled, differential equations describing heat and water transport (derived from Fourier's and Darcy's laws, respectively) in a one-dimensional soil profile (Jansson and Halldin, 1979). Snow dynamics, frost, evapotranspiration, precipitation, groundwater flow, plant water uptake and drainage flow are included. The model predicts soil climate variables (e.g. soil temperature, water content, etc.), with a daily resolution at any level in the soil profile. A detailed technical description of the model is given by Jansson and Halldin (1980), but a short summary is given here.

The model has a one-dimensional vertical structure and therefore lateral flow is not explicitly calculated. Water flow to drainage tiles occurs when the simulated groundwater is above the level of the tiles, i.e. flow occurs directly from a layer to drainage tiles when water content is at saturation. The flow rate is proportional to the hydraulic gradient (estimated from the depth of the groundwater table and the density of the drainage tiles) and the thickness and saturated conductivity of each layer. Calculation of natural drainage is based on an empirical assumption for net horizontal groundwater flow.

Surface runoff can occur because of a limited infiltration capacity or a limited hydraulic capacity of the soil. Surface runoff is assumed not to transport nitrogen. Runoff from surface and drainage flows are combined to give total runoff.

The water and heat model uses standard daily meteorological input data, i.e. air temperature, humidity, precipitation, global radiation (or duration of sunshine or cloud cover) and wind speed. If necessary, precipitation, air temperature and a simple estimate of potential evapotranspiration can suffice as input data. Parameter values for hydraulic and thermal soil properties can be estimated from standard soil physical characteristics, or independent measurements can be used.

The model, originally developed for a forest soil, has been modified and tested for agricultural soils (Jansson and Thoms-Hjärpe, 1986) and for application to agricultural watersheds (Lundin, 1984).

Nitrogen inputs

In the process descriptions given below, the following conventions have been used:

(1) Pools (state variables) are denoted by capitals subscripted with name abbreviations.

(2) Flows are denoted by capital italics subscripted with the direction of the transfer. Layer is indicated by "z" in parentheses.

(3) Parameters are indicated by lower-case italic letters with appropriate

subscripts. A list of parameters used in the nitrogen model is given in Appendix 1.

External input of nitrogen includes atmospheric deposition, inorganic fertilization and manure. Atmospheric deposition includes wet deposition, calculated from amount of precipitation and a mean mineral N concentration, and dry deposition, which is given a constant daily rate. Nitrogen fertilizer is added to the topsoil layer and a dissolution rate constant (e.g. for pelleted fertilizer) controls its input to the soil mineral N pools. Added manure is incorporated into the two uppermost soil layers. Manure additions are split into three fractions: (1) bedding (i.e. straw, sawdust, etc.) which is shunted to the soil litter pool; (2) faeces; (3) ammonium (Fig. 1).

Incorporation of plant material into the litter component is simulated twice: at harvest and when the field is ploughed. At harvest the amount of root N going to litter is

$$N_{p\rightarrow l}(z) = f_r(z)\ (1 - f_{ar} - f_{lr} - f_{hp})\ N_p \qquad (1)$$

where $f_r(z)$ is the fraction of roots in layer z, f_{ar} is the fraction of plant N remaining above ground after harvest, f_{lr} is the live root fraction of plant N after harvest, f_{hp} is the harvested fraction of plant and N_p is plant N prior to harvest. For an annual crop, f_{lr} would be 0 after harvest, while in a perennial crop, part or all of the root nitrogen could be excluded from transfer to the litter pool at harvest. Carbon flow to litter is proportional to nitrogen flow according to a constant C–N ratio for roots.

At ploughing, nitrogen in above-ground harvest residues (and remaining roots in a perennial crop) are incorporated into the plough layer.

$$N_{p\rightarrow l}(z) = (f_{ar} + f_{lr})\ N_p \qquad (2)$$

The carbon incorporated is proportional to eqn. (2) according to a constant C–N ratio for above-ground harvest residues (and roots for a perennial crop).

Mineralization, immobilization and nitrification

The general concept of active and passive components of soil organic matter is well established (Jansson, 1958; Stanford and Smith, 1972; Paul, 1984). Many models address the heterogeneity of organic material in soil by simulating multiple organic matter fractions (e.g. Paul and Juma, 1981; McGill et al., 1981; van Veen and Frissel, 1981; Parton et al., 1983). We have defined two principle organic matter pools: a fast cycling pool (litter), representing an organic matter–microbial biomass complex which receives fresh organic material, and a slow cycling pool (humus) composed of stabilized decomposition products (Fig. 1). A separate component was included for manure-derived faeces because of the substantial differences in its chemical composition compared with plant material entering the litter pool.

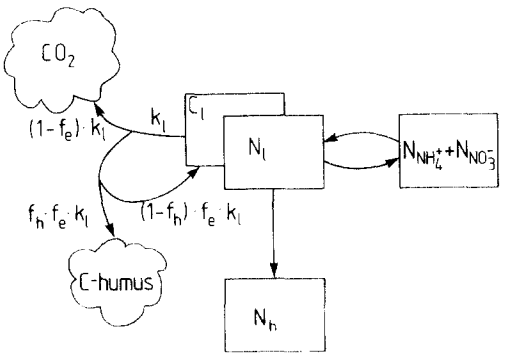


Fig. 2. Flow diagram showing carbon and nitrogen balance associated with litter (or faeces). Explanation of parameter symbols are given in the text.

Mineralization of humus ($N_h(z)$) is calculated as a first-order rate,

$$N_{h\rightarrow NH_4}(z) = k_h e_t(z)\ e_m(z)\ N_h(z) \tag{3}$$

where k_h is the specific mineralization constant and $e_t(z)$ and $e_m(z)$ are response functions for soil temperature and moisture, respectively (see below).

Decomposition in the two organic carbon pools (litter and faeces) are the main controls on N mineralization from these sources. Decomposition of soil litter carbon ($C_l(z)$)

$$C_{l(d)}(z) = k_l\ e_t(z)\ e_m(z)\ C_l(z) \tag{4}$$

is a function of a specific rate constant (k_l), temperature and moisture. The products of decomposition are CO_2, stabilized organic material (humus) and, conceptually, microbial biomass and metabolites (Fig. 2). Since a single pool (soil litter) represents the litter–decomposer complex, this synthesis of microbial biomass and metabolites constitutes an internal cycling (i.e. $C_{l\rightarrow l}(z)$, see below). The relative amounts of decomposition products formed,

$$C_{l\rightarrow CO_2}(z) = (1-f_e)\ C_{l(d)}(z) \tag{5}$$

$$C_{l\rightarrow h}(z) = f_e\ f_h\ C_{l(d)}(z) \tag{6}$$

and

$$C_{l\rightarrow l}(z) = f_e\ (1-f_h)\ C_{l(d)}(z) \tag{7}$$

are governed by a synthesis efficiency constant (f_e) and a humification fraction (f_h).

Mineralization or immobilization of nitrogen in the litter pool is governed by two assumptions: (1) that the internal cycling of carbon (eqn. 7) and the formation of humus (eqn. 6) have nitrogen demands determined by a constant C–N ratio of decomposer biomass and humification products (r_o); (2) that

nitrogen is released by decomposition of soil litter carbon in proportion to the actual litter C–N ratio. From eqns. (4), (6) and (7), net mineralization or immobilization of nitrogen in litter ($N_l(z)$)

$$N_{l\leftrightarrow NH_4}(z) = \left[\frac{N_l(z)}{C_l(z)} - \frac{f_e}{r_o} \right] C_{l(d)}(z) \tag{8}$$

is then determined by the balance between the release of nitrogen during decomposition and the nitrogen immobilized during microbial synthesis and humification. The switch between net immobilization and mineralization of N in litter occurs at the C–N ratio equal to r_o/f_e (cf. Parnas, 1975). A simplification is that the C–N ratio of organic matter humified is assumed to be the same as for microbial biomass and metabolites. Observed mean values of the C–N ratio of microorganisms and humus in agricultural soils are in the same range, normally between 5 and 15 (e.g. Parsons and Tinsley, 1975; Kowalenko, 1978; McGill et al., 1981).

When net immobilization occurs (i.e. $N_{l\leftrightarrow NH_4}(z) < 0$), immobilization will be reduced if sufficient mineral N is not available. The reduction in immobilization is determined in the same way as the reduction in plant uptake by roots (see eqn. 14), i.e. by assuming that a constant fraction of the total storage of mineral nitrogen is available. Both ammonium and nitrate can be immobilized, but with preference for available ammonium.

Nitrogen humified during litter decomposition is proportional to carbon humification (eqn. 6) according to the C–N ratio of microorganisms and humified products (r_o). Decomposition, mineralization/immobilization and humus formation for C and N in faeces are calculated in the same way as for litter.

Nitrification is not modelled explicitly as a microbial process, but the segregation of mineral N into ammonium and nitrate is made by assuming a nitrate–ammonium ratio characteristic for a particular soil. This simplification may be appropriate given rapid nitrification rates (e.g. Kowalenko, 1980; Berg and Rosswall, 1985) and the relatively small amount of ammonium often observed in agricultural soils (Cameron et al., 1978; Lindén, 1981). The transfer rate of ammonium to nitrate

$$N_{NH4\rightarrow NO_3}(z) = k_n\, e_t(z)\, e_m(z) \left[N_{NH_4}(z) - \frac{N_{NO_3}(z)}{n_q} \right] \tag{9}$$

depends on a potential rate (k_n) which is reduced as the nitrate–ammonium ratio (n_q) is approached. If $N_{NO_3}(z)/N_{NH_4}(z) > n_q$, no transfer of ammonium to nitrate occurs.

Abiotic response functions

Decomposition, mineralization and nitrification are regulated by the same abiotic response function involving soil temperature and moisture. A Q_{10} relationship (e.g. Bunnell et al., 1977) is used to express the effect of temperature,

$$e_t(z) = Q_{10}^{\left[\frac{T(z)-t_b}{10}\right]} \tag{10}$$

where $T(z)$ is the soil temperature for the layer, t_b is the base temperature at which $e_t(z)$ equals 1 and Q_{10} is the factor change in rate with a 10-degree change in temperature.

The soil moisture factor decreases, on either side of an optimum level, in drier soil or in excessively moist soil (Jansson and Berg, 1985), i.e.

$$e_m(z) = e_s + (1-e_s)\left[\frac{\theta_s(z)-\theta(z)}{\theta_s(z)-\theta_{ho}(z)}\right]^m \quad \theta_s(z) \geq \theta(z) > \theta_{ho}(z) \tag{11a}$$

$$e_m(z) = 1 \quad \theta_{ho}(z) \geq \theta(z) \geq \theta_{lo}(z) \tag{11b}$$

$$e_m(z) = \left[\frac{\theta(z)-\theta_w(z)}{\theta_{lo}(z)-\theta_w(z)}\right]^m \quad \theta_{lo}(z) > \theta(z) \geq \theta_w(z) \tag{11c}$$

where $\theta_s(z)$ is the saturated water content, $\theta_{ho}(z)$ and $\theta_{lo}(z)$ are the high and low water contents, respectively, for which the soil moisture factor is optimal, and $\theta_w(z)$ is the minimum water content for process activity. A coefficient (e_s) defines the relative effect of moisture when the soil is completely saturated and m is an empirical constant. The two thresholds, defining the optimal range, are calculated as

$$\theta_{lo}(z) = \theta_w(z) + \Delta\theta_1 \tag{12a}$$

$$\theta_{ho}(z) = \theta_s(z) - \Delta\theta_2 \tag{12b}$$

where $\Delta\theta_1$ is the volumetric range of water content where the response increases and $\Delta\theta_2$ is the corresponding range where the response decreases.

Plant uptake of nitrogen

Plant uptake of N includes both nitrate and ammonium from each layer where roots are present. A logistic uptake curve (cf. Greenwood et al., 1974) is used to define the cumulative potential N demand during the growing season,

$$\int u(t)\,dt = \frac{u_a}{1+\frac{u_a-u_b}{u_b}e^{-u_c t}} \tag{13}$$

where u_a is the potential annual N uptake, u_b and u_c are shape parameters and t is days after the start of the growing season. Diffusion and root absorption are assumed to be the main mechanisms behind nitrogen uptake by roots (Nye, 1977). Consequently, the uptake is not associated with plant water uptake. To limit uptake if soil mineral N concentration is low, a maximum availability

fraction (f_{ma}) is assumed which is proportional to the total amount of ammonium and nitrate in each layer. Daily uptake of nitrate

$$N_{NO_3\text{-}p}(z) - MIN \left| \begin{array}{ll} f_r(z) \dfrac{N_{NO_3}(z)}{N_{NO_3}(z) + N_{NH_4}(z)} u & (14a) \\ f_{ma}\, N_{NO_3}(z) & (14b) \end{array} \right.$$

is then calculated from the relative root fraction in the layer ($f_r(z)$), the proportion of total mineral N as nitrate and the derivative of the growth curve (u). An empirical root distribution, which changes during the growing season, controls relative uptake rates from each layer. Actual uptake is taken as the minimum of the two functions. Ammonium uptake is calculated in the same way, but is proportional to the relative amount of ammonium in total mineral N. If actual uptake from a layer is below the potential rate, compensatory increases in uptake may occur from other layers (Drew and Saker, 1975). This is achieved by adding the difference between potential and actual uptake rates to the potential demand for the soil layer below.

Denitrification

Denitrification, the reduction of nitrate to gaseous nitrogen (N_2 and N_2O) products, is an anaerobic process and consequently is highly dependent on soil aeration. The model uses soil water content ($\theta(z)$) as an indirect expression of soil oxygen status, where the influence on the denitrification rate

$$e_{md}(z) = \left[\frac{\theta(z) - \theta_d(z)}{\theta_s(z) - \theta_d(z)} \right]^d \tag{15}$$

is expressed as a power function which increases from a threshold point ($\theta_d(z)$) and is maximum at saturation ($\theta_s(z)$), where d is an empirical constant. Below the threshold water-content no denitrification occurs. This expression is similar to the soil-water relationship developed by Rolston et al. (1984). The denitrification rate for each layer

$$N_{NO_3\text{-}}(z) = k_d(z)\, e_{md}(z)\, e_t(z) \left[\frac{[N_{NO_3}(z)]}{[N_{NO_3}(z)] + c_s} \right] \tag{16}$$

depends on a potential denitrification rate ($k_d(z)$), the soil water/aeration status ($e_{md}(z)$) and the same temperature factor ($e_t(z)$) used for the other biologically-controlled processes. The effect of nitrate concentration is controlled by the half-saturation constant, c_s (i.e. the concentration where the rate is 50% of the maximum, if all other conditions are optimal).

Nitrate transport and leaching

Nitrate is considered to be wholly in solution and movement between layers and leaching from the profile in drainage flow are determined by water flow

rates. The nitrate transport is thus calculated as the product of the water flow and the nitrate concentration in the soil layer from which the water flow originates.

Ammonium is assumed to be immobile with respect to leaching (i.e. largely fixed to soil colloids) and is not transported by water flow (Krantz et al., 1943; Broadbent et al., 1958; Wiklander, 1977).

MODEL APPLICATION

Site description

Kjettslinge was the experimental site for the project "Ecology of Arable Land". In the project, data were collected on decomposition, denitrification, leaching, mineralization, mineral N profiles, primary production and soil abiotic conditions. A detailed description of the field site is found in Steen et al. (1984) and only a brief summary of site characteristics is given here.

The field site is situated in central Sweden, approximately 40 km north of Uppsala. The climate in the region is cold-temperate and semi-humid. Annual precipitation and mean annual temperatures at Ultuna, 10 km south of Uppsala, are 520 mm and 5.4°C (1931–60), respectively (Rodskjer and Tuvesson, 1975). The experimental field has two parts: the main field and four tile-drained plots used for leaching studies. The cropping systems chosen for the simulations were (1) B0, barley with no N fertilization, and (2) B120, barley receiving 120 kg N ha^{-1} $year^{-1}$ as calcium nitrate fertilizer.

Data on leaching (Bergström, 1987), soil mineral N profiles (Bergström, 1986) and crop production (R. Pettersson, personal communication, 1985) were taken from the tile-drained plots, which were established in 1981, following a fallow year in 1980.

The soil, down to a depth of 1.0 m, consists of four distinct layers of varying texture and structure: (1) topsoil, consisting of clay loam, with a mean thickness of 0.27 m; (2) a fine sand layer, with a thickness varying from 0 to 0.50 m; (3) an oxidized clay layer, down to a depth of approximately 0.75 m; (4) a non-oxidized clay layer, below the oxidized clay layer.

The topsoil consists of 15–20% clay, 2.2% organic carbon and 0.23% organic nitrogen and has a pH of 6.0–6.5.

The soil water and heat simulation

Simulation of soil water and heat processes, which are inputs to the nitrogen model, were based on a previous application to the field (Jansson and Thoms-Hjärpe, 1986).

The meteorological data used as driving variables were measured at the experimental field (Alvenäs et al., 1986). Missing data were substituted with

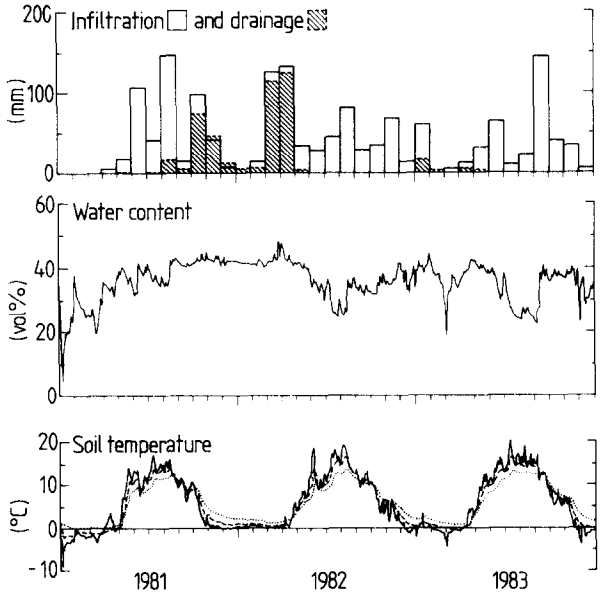


Fig. 3. Simulated abiotic conditions at Kjettslinge: monthly values of drainage, infiltration and daily mean values of soil water content in the topsoil and daily values of soil temperature at depths of 0.05, 0.36 and 0.85 m.

data from the meteorological stations at Marsta and at F16, both ca. 30 km south of the Kjettslinge field (Alvenäs et al., 1986; Jansson and Thoms-Hjärpe, 1986).

The soil profiles in both the soil-water and heat model and in the nitrogen model were divided into 5 layers according to the textural horizons in the field and numerical demands in the water and heat model (Appendix 1). A common set of soil physical parameters was used for the tile-drained plots, as opposed to the earlier application by Jansson and Thoms-Hjärpe (1986), which used separate parameter sets for each cropping system in the main field. Because of a lack of data on soil moisture dynamics in the tile-drained plots and since the model was concentrated on biological processes, this simplification was deemed appropriate.

Drainage was assumed to take place only through the drainage tiles. Measurements of groundwater pressure at the site showed a slight tendency for a positive pressure gradient from the subsoil (Bergström, 1987), indicating stable conditions or an inflow of water to the profile.

Simulated soil temperatures, infiltration, drainage and soil water content showed substantial differences between years (Fig. 3). Two summers, 1982 and 1983, were characterized by rather high temperatures with extended drought periods, whereas the summer/autumn of 1981 was comparatively wet.

A thick snow cover prevented the soil from becoming completely frozen during the winter of 1981/82.

The simulated drainage flow was adjusted, compared with that used by Jansson and Thoms-Hjärpe (1986), to achieve better agreement with drainage measurements from the B120 plot (Bergström, 1986a). The adjustment was done by changing evapotranspiration parameters controlling the water balance in the model.

Parameter derivation

The adaptation of the nitrogen model to the two barley treatments (B0 and B120) at the Kjettslinge field was done in two steps. First, parameter values were estimated from independent measurements at the field or estimated from the literature. In a second step, a few parameter values, which were found to strongly influence the results, were adjusted to improve the agreement between measured and simulated mineral N profiles. Adjusted parameters were the nitrogen availability fraction, litter and humus turnover rates, Q_{10}-value and saturation activity (Appendix 1). This second step was restricted to model and data comparisons for the B120-treatment. Subsequent simulations of the B0-treatment were not adjusted and the only difference was the lack of application of N-fertilizer. The parameter values chosen are summarized in Appendix 1.

Mean value for dry deposition of mineral nitrogen was set to 0.001 g N m^{-2} day^{-1} (Monitor, 1984). To simulate wet deposition a N concentration of 0.8 mg l^{-1} in precipitation was used (Monitor, 1984).

Parameter estimates for organic matter flows were based on data from the field site and from literature values. The microbial growth yield and microbial C–N ratio, parameters associated with the litter pool, were set to 0.5 (Kjøller and Struwe, 1982) and 10 (McGill et al., 1981), respectively. These together determine a critical C–N ratio for net N mineralization of 20. Litter decomposition rates were based on litter bag incubations (Andrén and Paustian, 1987) and we assumed that 10% of decomposing litter carbon became stabilized (cf. Jenkinson, 1977) and entered the humus pool.

A Q_{10}-value of 3 was used for the effect of temperature on decomposition. Reported values for Q_{10} generally range between 2 and 3 in temperate soil. For example, Campbell et al. (1984) found mean Q_{10} values, for mineralization, of 2.0, 2.18 and 2.75 for three major soil groups in Canada. In general, Q_{10} values tend to be larger for lower temperature intervals. In this study, with temperatures often around freezing during long periods, a relatively high Q_{10}-value was used to prevent unreasonably high activity rates at low temperatures.

The influence of soil moisture was expressed in straight-line segments as a function of unfrozen soil water content. The function had a value of 0 at the wilting point for each layer, increasing to 1 at a volumetric water content of

10% above the wilting point. For water contents greater than 8% below saturation, the value again declined to 0.6 at saturation. The soil moisture function also accounted for frost effects below 0 °C, because of water freezing.

Simulated plant uptake started 2 weeks after sowing (20 May) and ended at harvest (25 August), with the maximum uptake rate in late June. The same dates for sowing plus fertilization and for harvest were used each year in the simulation and they represented means of the actual dates for the 3 years. Potential net nitrogen uptake was based on N-biomass measurements (including roots) from the field (Hansson et al., 1987; R. Pettersson, personal communication, 1985). Measurements of root production in N-fertilized and unfertilized barley indicated that 20–40% of plant nitrogen was found in the roots (Hansson et al., 1987), depending on the estimation method, stage of crop development and treatment. We assumed 30% of total plant nitrogen to be allocated to the roots. At harvest, 60% of nitrogen in plant biomass was removed and the 30% in the roots was transferred to the litter component. Harvest residues, 10% of total N, were incorporated into the litter pool in soil at ploughing. The C–N ratios assumed for roots (25) and for harvest residues (50) were based on mean values of chemical data for crop biomass during 1981 (Hansson et al., 1987).

For calculating nitrate uptake from different levels in the profile, changes in root depth distribution over time were based on field data (Hansson et al., 1987) for both crops. However, daily uptake from each layer was set to a maximum of 8% of the total mineral N content in the layer if potential uptake exceeded this value.

The potential rate of denitrification was estimated to be 0.1 g N m^{-2} day^{-1} (L. Klemedtsson and B. Svensson, personal communication, 1985). The actual rate was reduced to zero when the air-filled pore space exceeded a volumetric percentage of 10, but increased rapidly when the soil approached saturation (cf. Rolston et al., 1984).

RESULTS AND DISCUSSION

Variations in the soil mineral N content

The overall seasonal dynamics of simulated mineral N content of the soil and the differences between years, i.e. for different climatic conditions, agreed well with observed data (Fig. 4). For example, the model showed the decrease in nitrate content in the B120 profile during the wet autumn of 1981, as opposed to the increases in nitrate in the drier autumns of 1982 and 1983 (Fig. 4).

Agreement with observed values were better for surface horizons (0–27 and 27–45 cm) in both B120 and B0 (Fig. 4) than for deeper horizons. Simulated nitrate levels in the two clay layers were usually more static than the observed

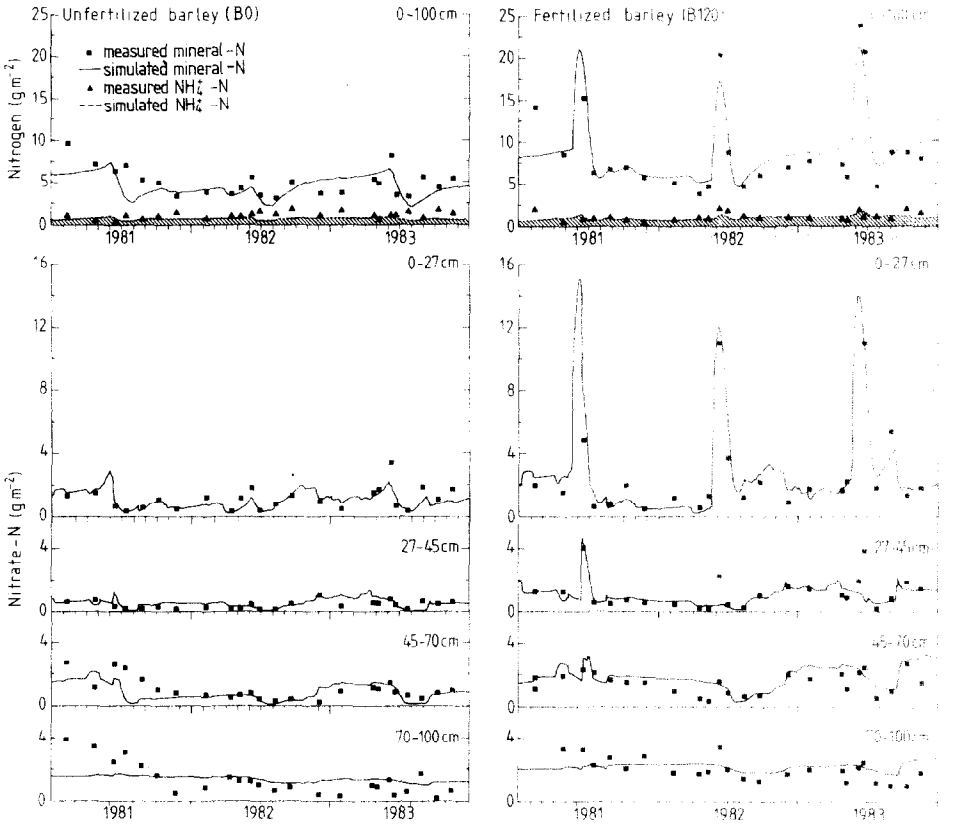


Fig. 4. Simulated and measured storage of mineral N, ammonium N and nitrate N in unfertilized barley and in N-fertilized barley at Kjettslinge.

levels. This appeared to be due to difficulties in simulating nitrogen transport and water flow between subsurface layers in combination with the flow to drainage tiles from different layers. For example, it was not possible to simulate the large decrease in nitrate in spring 1981, which had accumulated in the lower layers following the fallow in 1980. The initial nitrate levels were therefore set to the values obtained at the second sampling date (Fig. 4). An underestimation of water flow from the surface layer to the sand (25–45 cm) may also be the reason for the missing peaks in nitrate amounts found in the sand layer following fertilization (Fig. 4).

A considerable deviation from observed data occurred when the model did not show the sudden drop in nitrate content just prior to N-fertilization in May 1983 (Fig. 4). This decrease was observed in the two clay layers, but almost no leaching was observed in the drainage flow. However, leaching was simulated by the model and most of the water flow came from the sand layer. In

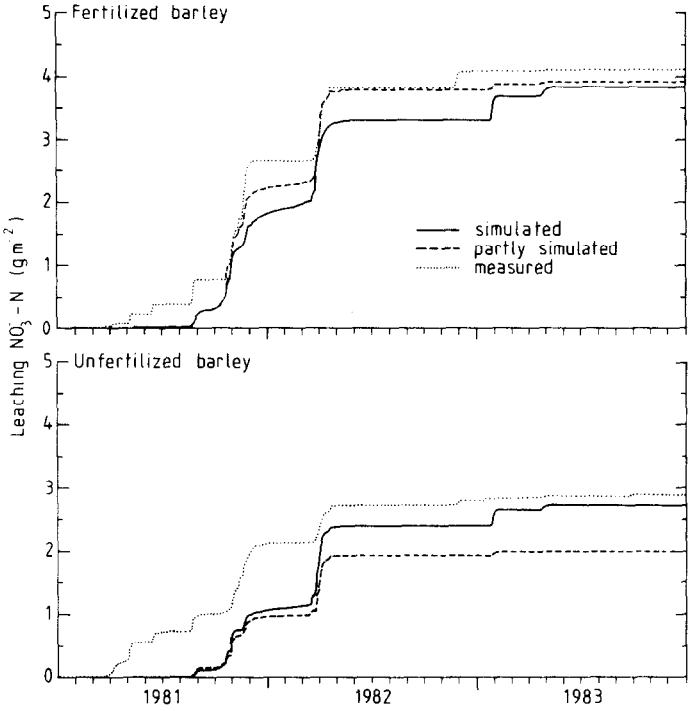


Fig. 5. Simulated, partly simulated (measured nitrate concentrations multiplied by simulated water flows from drainage tiles) and measured leaching from N-fertilized barley and unfertilized barley at Kjettslinge.

this case, measurements of groundwater pressure (Bergström, 1987) indicated that the groundwater table was below the drainage tiles during the spring of 1983, suggesting that percolating water reached the groundwater table rather than being discharged through the drainage system. The observed nitrate levels in the clay layers showed considerable dynamics. This indicated a greater degree of water movement in the lower profile than was predicted by the hydrological simulations.

Some of the deviations in mineral N levels for the two upper layers may be attributed to the simulation of mineralization. For example, mineralization appeared to be underestimated in the autumn of 1981 and 1983 in B120 and in late spring of 1982 and 1983, after fertilization, in B0 and B120 (Fig. 4). Simulated mineralization rates were higher in the fertilized barley because of greater crop residue incorporation (Table I). Simulated litter decomposition resulted in a net release of mineral N, although about 50% of litter N inputs to B120 and slightly less in B0 were stabilized in the humus fraction (Table I).

The empirical uptake equation was satisfactory in predicting the uptake of both crops (Table I; R. Pettersson, personal communication, 1985) in relation to mineral N availability. Availability of mineral N was the only factor limiting crop growth and thus causing differences between the B0- and the B120-treat-

TABLE I

Annual means of simulated nitrogen flows based on the 3-year simulations. Partial flows given in parentheses. All values in g N m^{-2} $year^{-1}$

Flow	N-fertilized barley (B120)	Unfertilized barley (B0)
Fertilization and deposition	12.8	0.8
Mineralization	8.0	6.9
Net litter mineralization	(3.4)	(2.3)
Humus mineralization	(4.6)	(4.6)
Humification	3.3	1.7
Plant uptake	18.3	6.8
Crop residues to litter	(7.3)	(2.7)
Harvest	(11.0)	(4.1)
Leaching	1.3	0.9
Denitrification	0.4	0.4

ment. Since crop uptake is normally the dominant N flow in cropped systems, small relative changes, especially in root distribution and nitrogen availability, in the simulation of crop uptake are critical for the prediction of soil nitrate levels.

The variation of mineral N in the soil was mainly due to nitrate dynamics. The ammonium level, simulated and measured, was quite stable and similar in both treatments.

Nitrogen losses

The simulation of nitrate leaching was dependent on two model outputs: soil nitrate concentration and drainage water flow. In addition to observed leaching calculated from data on drainage water flow and nitrate concentrations, time series calculated from measured nitrate concentrations and simulated drainage (i.e. "partly simulated", Fig. 5) were compared to model predictions (i.e. "simulated", Fig. 5). The partly simulated time series was included to isolate deviations in leaching patterns due to either poor prediction of drainage water flows or of nitrate levels.

Cumulative nitrate leaching was similar in all three time series (Fig. 5). Due to an underestimate in drainage water flow, the model failed to simulate the leaching which occurred in both treatments during the first half of 1981. After this time-period, drainage flows were well predicted by the model as indicated by the parallel courses of the partly simulated and measured time series.

The model overestimated nitrate concentrations in the drainage water dur-

ing the spring of 1982 and 1983 in B0 and in spring 1983 in B120 (Fig. 5). In B0, simulated drainage from upper soil layers was too high and redistribution of water between layers was underestimated. In B120, leaching was observed during the previous autumn (Fig. 5) but was not simulated by the model, which may explain the overestimation of nitrate concentrations (and loss) for the following spring leaching period. The overall effect was a lag in the simulated leaching, but the cumulative amounts at the end of the period were similar. Overestimates of leaching on these occasions (Fig. 5) may indicate deviations from the assumed flow paths, e.g. percolation to groundwater instead of to drainage tiles (cf. Bergström, 1987).

Underestimations of simulated vs. observed nitrate concentrations were most evident in autumn 1981 in fertilized barley. Here the underestimation in mineralization rates, discussed above, may explain the differences.

Simulated N losses due to denitrification were less than one-half of leaching losses in both treatments (Table I). The yearly totals of denitrification were similar to preliminary estimates, based on field measurements at Kjettslinge, of ca. 0.5 g N m^{-2} $year^{-1}$ in B120 (Klemedtsson, 1986). This suggests that there were no large discrepancies, in other comparisons of data and model predictions, due to unknown losses via denitrification. An evaluation of the simulation of temporal dynamics of denitrification, in comparison with field measurements (Klemedtsson, 1986), will be presented elsewhere.

CONCLUSIONS

The present application showed generally good agreement between simulations and measurements. Both abiotic and biotic factors were found to be important when analyzing the dynamic behaviour of mineral N in the system. Disagreements between simulated and measured mineral N contents in the topsoil may be explained by biotic factors, whereas disagreements in deeper layers may be explained by abiotic factors.

The simulated nitrate dynamics in the topsoil showed patterns similar to the measured values in both the B0- and B120-treatments. However, the model often predicted a more dampened seasonal variation in nitrate levels, especially for the B0-treatment. This may be due to an inadequate description of temperature and moisture influences, including possible stimulus of activity from wet–dry and freeze–thaw cycles on decomposition and mineralization (Soulides and Allison, 1961). Also, to account for spring and autumn mineralization peaks, it may be necessary to include a labile pool of easily decomposable, nitrogen-rich material in crop residues (McGill et al., 1981).

Spatial relationships between root distribution and nitrate uptake from different levels of the profile are not well understood (Nye and Tinker, 1977). Our preliminary approach to modelling uptake showed a strong sensitivity to the chosen root distribution with depth. Static information on root biomass

distributions may be of minor value because of compensatory growth and uptake by roots (Drew and Saker, 1975). A new formulation of this process in the model should more strongly emphasize the dynamic nature of the roots and root uptake activity.

The use of the fixed potential uptake curve to control uptake is a simple approach which mainly requires information on expected crop growth. However, it assumes that nitrogen availability is the only limiting factor for uptake and, currently, no consideration of other time-varying factors such as drought stress or meteorological elements are included. Also the definition of potential uptake is fairly specific to crop and site.

Leaching was simulated mainly from the sand layer and upper clay layer (45–70 cm). Measurements of mineral N in the two clay layers (45–100 cm) indicated that the main flow path was probably deeper. Both the water flow rates and the flow paths were shown to strongly influence the temporal patterns of leaching.

Leaching was not only dependent on the water flow rate but also on the flow path and the vertical distribution of nitrate in the soil profile. The measured and simulated depth distributions of nitrate showed that a layered model structure improves simulation of nitrate leaching. The simplifying assumption that nitrate was only associated with mass flow, i.e. diffusion and dispersion could be ignored, appeared reasonable.

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APPENDIX 1

Parameters in the nitrogen model and values used for the application

Parameter definition	Equation	Symbol	Unit	Value
External inputs				
N-fertilization			$g\ N\ m^{-2}\ year^{-1}$	0/12[1]
Dry deposition			$g\ N\ m^{-2}\ day^{-1}$	0.001
Wet deposition, concentration			$mg\ l^{-1}$	0.8
Fertilizer specific dissolution rate			day^{-1}	0.15
Crop and management				
Fertilization date				20 May
Harvest date				25 Aug.
Ploughing date	(1)	f_{hp}	—	16 Oct.
Harvested fraction	(1)	f_{ar}	—	0.6
Above ground residue fraction	(1)	f_{lr}	—	0.1
Live root fraction				0
C–N ratio of above-ground residues			—	50
C–N ratio of roots			—	25
Ploughing depth			m	0.27
Mineralization and immobilization				
Humus specific mineralization rate	(3)	k_h	day^{-1}	7.0×10^{-5}
Litter specific decomposition rate	(4)	k_l	day^{-1}	0.035

Parameter definition	Equation	Symbol	Unit	Value
Efficiency	(5)	f_e	—	0.5
Carbon humification fraction	(6)	f_h	—	0.2
C–N ratio of microorganisms and humified products	(8)	r_o	—	10
Fraction of available mineral N			day^{-1}	0.08
Specific nitrification rate	(9)	k_n	day^{-1}	0.2
Nitrate–ammonium ratio	(9)	n_q	—	8
Soil moisture response				
Saturation activity	(11)	e_s	—	0.6
Increasing interval	(12)	$\Delta\theta_1$	%	10
Decreasing interval	(12)	$\Delta\theta_2$	%	8
Porosity	(11)	$\theta_s(z)$	%	45/45/40/53/53[2]
Wilting point	(11)	$\theta_w(z)$	%	15/15/4/27/27[2]
Coefficient	(11)	m	—	1
Soil temperature response				
Response to a 10 °C change	(10)	Q_{10}	—	3
Optimal temperature	(10)	t_b	°C	20

Plant uptake				
Start of plant uptake				3 June
Potential nitrogen uptake	(13)	u_a	g N m^{-2} $year^{-1}$	20
Coefficient	(13)	u_b	—	1
Coefficient	(13)	u_c	day^{-1}	0.12
Fraction of available mineral N	(14)	f_{ma}	day^{-1}	0.08
First date of roots in layer z				3/12/20/29/-June[2]
Fraction of roots in layer z (when fully developed)			—	0.6/0.28/0.08/0.04/0[2]
Denitrification				
Potential rate	(16)	$k_d(z)$	g N m^{-2} day^{-1}	0.1
Half-saturation constant	(16)	c_s	mg l^{-1}	10
Soil moisture effect:				
Activity range			%	10
Coefficient	(15)	d	—	2
Structural				
Thickness of soil layer z			m	0.10/0.17/0.18/0.25/0.30[2]

[1]Unfertilized and fertilized barley.
[2]For layers 1–5.

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